

2019 J2 H2 Physics: Preliminary Examination (Paper 3) Solutions

1. (a) The pressure due to a fluid increases with depth in the fluid. [B1]
For a body in a fluid, there is a difference in fluid pressure over the upper and lower surface of a body, which gives rise to upthrust on the body. [B1]

- (b) (i) Consider the system of object and balloon.

Total upthrust – total weight = total mass x acceleration

$$1000g(V_o + 0.15) - 305.2g = 305.2a \quad [\text{C1}]$$

$$V_o = 0.217 \text{ m}^3 \quad [\text{A1}]$$

- (i) 1. A body moving in a fluid experiences a drag force which opposes its motion. [B1]
This drag force is proportional to the velocity of the body Or increases as the velocity increases [B1] The drag force causes the net force to decrease, hence acceleration decreases continuously.

- (ii) 2. As the lifting balloon rises, the fluid pressure decreases which causes the volume of the balloon to increase [B1]. Upthrust on the bag thus increases [B1], leading to an increase in net force and acceleration.